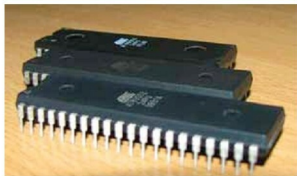




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The 8051 microcontroller architecture

have a big say on the system processing needs. Based on these requirements, you select the MCU. The question is: "Should I choose strictly based on the requirements from a commercial perspective or consider the greater good of all?"

"This part is usually taken very seriously during MCU selection in our firm. We usually do not prefer to use too many different types of MCUs in our products only to tightly match the application requirements. This would leave us with too many tools of development and maintenance, which unnecessarily blocks lots of money. Instead, we have carefully chosen a few powerful devices and also a few low-profile devices that we choose from," explains T. Anand, managing director, Knewron.

Plato Pathrose, senior engineer, design & development, Tata Elxsi, explains how things work in the automotive industry: "Automotive compliance of the MCU should be considered at first; without this, nothing could be taken inside the automotive industry. Next, we look for product familiarity in development, which could reduce the development costs and increase the flexibility in usage. Following this, we look into the availability of the tools and samples and also the vendor's support during the

course of development."

"The key parameters to be considered while selecting an MCU should also include the possibility of reduction in the number of electronic components with the MCU. This will allow integration of more functionality. For instance, you can integrate more memory (Flash/RAM) and a multiprocessor," adds Prashant Mahendrakar, technical manager at Sankalp Semiconductor.

Don't forget about memory

The cutthroat competition in the market has pushed software development so much that we now live in a world that seems to be obsessed with the 'Beta Culture.' Almost every new service or product is run on Beta software. The designer should keep this in mind while choosing the memory size for his product. The software that he has been given now may soon bloat up as bugs get squashed and new features added. So there better be memory available for the update.

Jigar Patel says, "Some algorithms require substantial RAM to be implemented in a straightforward manner. It may be worthwhile looking for a micro with a lot of RAM. On

the other hand, the size of the Flash memory depends on the size of the object file generated by the assembler/compiler. The size of the Flash program memory is an important factor when the user has decided to use the internal program memory. The user must keep in mind to keep a few percentage (depending on the nature of the application) of the free space in the program memory for future requirements."

Anand adds, "When choosing memory sizes (of all kinds), we give a thought to scalability (if it is planned) such that tomorrow we may update the program; this would need additional memory, so the installed MCU should have the margin already built-in. This margin has to be around 15-20 per cent. MCU sizes usually come in different slots such as 4kB, 8kB, 16kB and so on. Therefore we go with the closest possible option. When in doubt, we go with one slot higher option."

Scalability is very important for a commercial embedded system to stay viable in the market over the long run.

Shrenik Shikhare, embedded software engineer, Maven Systems, says, "Embedded designers must have their eye beyond their designs and think strategically when selecting platforms with appropriate migration paths for long duration, feature set and performance. As scalability is the ability to handle the growing workload that may come in the future, one should think in all directions — whether the system will perform well to handle the extra work after adding some hardware into the existing system, or the existing



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